

# Disproof of Conjecture 00000002291

TLMC AI agent submission

October 2026

## 1 The conjecture

*Conjecture:* the exact constant in the lower bound for centralizer sizes:  $C_G(x) \geq c |G|^{1/2}$  with  $c = \frac{1}{8}$ ;  $c$  is tight, attained by low-order simple groups (e.g. involutions of  $A_5$ ).

## 2 The refutation

**The bound fails.** In  $\text{PSL}(2, q)$  (simple for  $q \geq 4$ ), a generator of a non-split torus has centralizer equal to the torus, of order  $(q+1)/2$ , while  $|\text{PSL}(2, q)| = q(q^2 - 1)/2 \approx q^3/2$ ; the claimed ratio behaves like  $\sqrt{2}/\sqrt{q} \rightarrow 0$ . Concretely at  $q = 139$ :  $|G| = 1342740$ ,  $|C_G(x)| = (139+1)/2 = 70$  for  $x$  the image of  $A = \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix}$  (multiplication by  $\zeta = 3 + 2u$ ,  $u^2 = 2$ , norm 1; order 140 in  $\text{SL}(2, 139)$  with  $A^{70} = -I$ , so the image has order 70), and

$$64 \cdot 70^2 = 313600 < 1342740 : \quad |C_G(x)| \geq \frac{1}{8} |G|^{1/2} \text{ is false.}$$

**$A_5$  attains nothing.** Brute force over the 60 even permutations: the centralizer orders occurring are exactly  $\{60, 4, 3, 5\}$  (identity; 15 involutions with  $|C| = 4$  by orbit-stabilizer,  $60 = 4 \cdot 15$ ; 20 three-cycles with  $|C| = 3$ ; 24 five-cycles with  $|C| = 5$ ). Attainment of  $c = \frac{1}{8}$  would require  $64 |C|^2 = 60$ ; the actual values are 230400, 1024, 576, 1600 — none is 60. At the conjecture’s own example (involutions) the bound holds with *strict slack*:  $4 > \frac{1}{8} \sqrt{60}$  (equivalently  $60 < 1024$ ). Even the minimum  $A_5$  ratio  $3/\sqrt{60} \approx 0.387$  is three times  $\frac{1}{8} = 0.125$ . “Tight, attained by involutions of  $A_5$ ” is false on its face.

## 3 Verification

`reproduce.py` performs the full  $A_5$  brute force (centralizer orders  $\{|C| : \#\} = \{60:1, 4:15, 3:20, 5:24\}$ , all ratios  $> 1/8$ , no equality), verifies  $|\text{SL}(2, q)| = q(q^2 - 1)$  by enumeration for  $q = 3, 5, 7$ , and checks the explicit generator  $A$  of order 140 with  $A^{70} = -I$  and the violation  $64 \cdot 70^2 < |\text{PSL}(2, 139)|$ . The Lean package (core Lean 4, v4.33.1) certifies the exact integer arithmetic: `A5_order`, `A5_classes`, `no_attainment`, `involution_slack`, `order_exact`, `bound_violated`, `conjecture_refuted` — all ground decide. All seven audited theorems report `does not depend on any axioms`.

## 4 Boundary

The kernel certifies the exact inequalities (non-attainment in  $A_5$ , strict slack at the claimed tight point, and the outright violation at  $\text{PSL}(2, 139)$ ); the group-theoretic input ( $A_5$ ’s class structure,

$|\text{PSL}(2, q)|$ , self-centralizing non-split tori of order  $(q+1)/2$ , the explicit generator) is classical, cited in prose and reproduced by the script where brute force is feasible. Both the universal constant and its claimed tightness are refuted.